traversed, because we need to know what we can return. Since backtracking walks the path backwards, the first key node we find while backtracking will hold the longest prefix.

Listing 6.9 method longestPrefix

```
function longestPrefix(node, s,
prefix)                                          #1
  if s == “” then
                                                      #2
    if node.keyNode
then                                                        #3
      return prefix
    else
      return
null
  #4
  c, tail ←
s.splitAt(0)
   #5
  if node.children[c] == null then
                          #6
    return null
  else
    result ← search(node.children[c], tail, prefix +
c)                        #7
    if result != null
then                                                  #8
      return result
    elsif node.keyNode
then                                                 #9
      return prefix
    else
      return
null
  #10
```